We calculate $[\boldsymbol{f},[\boldsymbol{f},[\boldsymbol{f},\boldsymbol{g}]]]$ and project onto the space spanned by the first three components to find

$$\mathbf{u}_3 = \sigma\left(\frac{2\gamma p^3 - 4p^3\xi_2 + 4p^2 q + \xi_1^2 p\xi_2 + \gamma\xi_1^2 p - 2\xi_1^2 q}{D^2}\right)\partial_q$$

$$-\sigma\xi_1^2\left(\frac{\eta}{D^3}\right)\partial_p + 2\sigma\xi_1 p\left(\frac{\eta}{D^3}\right)\partial_{\xi_1}. \tag{8.54}$$

with

$$\begin{aligned}\eta = {} & 4p^5\gamma^2 - 8p^5\xi_1^2 - 4\xi_1^2 p^3 + 4p^3\gamma^2\xi_1^2 - 4p^3\xi_1^4 \\ & - 2\gamma\xi_1^2 p^2 q + 4p^2 q\xi_2\xi_1^2 - 4\xi_1^2 q^2 p - 2\xi_1^4 p + p\gamma^2\xi_1^4 - q\gamma\xi_1^4 + \xi_1^4\xi_2 q.\end{aligned} \tag{8.55}$$

Along $p = 0$, $\mathbf{u}_3$ becomes

$$\mathbf{u}_3|_{p=0} = -2\sigma q\partial_q - \sigma\left(\frac{-q\gamma\xi_1^4 + \xi_1^4\xi_2 q}{\xi_1^4}\right)\partial_p$$

$$= -2\sigma q\partial_q - \sigma q(-\gamma + \xi_2)\partial_p. \tag{8.56}$$

Together with $\mathbf{u}_0$ we obtain a linearly independent set as long as $q \neq 0$.

If $p = 0$ and $q = 0$ then we are on $\mathcal{M}_0$. If $p \neq 0$ then $\boldsymbol{u}_2$ and $\boldsymbol{u}_0$ are linearly independent. If $q \neq 0$ then $\boldsymbol{u}_3$ and $\boldsymbol{u}_0$ are linearly independent. From these we establish that the Hörmander condition holds everywhere except on a space of co-dimension 2 (that is except on the one-dimensional set $\mathcal{M}_0$).

In this case, it is possible to complete the proof by introducing a Lyapunov function ϕ to demonstrate the necessary contractivity in the ξ_2 direction. The generator of this system is

$$\mathcal{L}_{\text{isok}} = p\frac{\partial}{\partial q} + \left[F - (p^2/\Lambda)F\right]\frac{\partial}{\partial p} - \frac{p\xi_1 F}{\Lambda}\frac{\partial}{\partial\xi_1} + \mathcal{L}_{\text{N}} + \mathcal{L}_{\text{O}}$$

with $\Lambda = p^2 + \xi_1^2/2$,

$$\mathcal{L}_{\text{N}} = \frac{1}{2}\frac{\xi_1^2\xi_2 p}{\Lambda}\frac{\partial}{\partial p} + \frac{1}{2}\frac{\xi_1^3\xi_2}{\Lambda}\frac{\partial}{\partial\xi_1} - \xi_2\xi_1\frac{\partial}{\partial\xi_1} + [\xi_1^2 - \beta^{-1}]\frac{\partial}{\partial\xi_2},$$

and

$$\mathcal{L}_{\text{O}} = -\gamma\xi_2\frac{\partial}{\partial\xi_2} + \frac{\sigma^2}{2}\frac{\partial^2}{\partial\xi_2^2}.$$